EDUX / ACADEMIC INTELLIGENCE

FULL EXAM ANALYSIS REPORT

QA Three-Subject Fixture — extensive revision priorities and mathematical solutions

JEE Main 2027 · Full Syllabus Mock · JEE Main (Paper 1) · All Subjects · Generated 02 Nov 2026

STUDENT	ROLL NUMBER	EXAMINATION DATE	TOTAL MARKS
Aarav Sharma	21CS114	2026-08-01	360

TOTAL MARKS OBTAINED

244

/ 360 marks

61 correct | 23 incorrect | 6 skipped





Correct 61 · Incorrect 23 · Skipped 6

ACCURACY	PERCENTILE	RANK
72.6%	94.2	#1260

PUBLISHED PERFORMANCE BRIEF

Synthetic three-subject fixture with a deliberately large published revision list and long mathematical explanations. It is not a student report.

01 PERFORMANCE INTELLIGENCE

Performance intelligence

Response composition

90 questions in the paper



- Correct 61
- Incorrect 23
- Skipped 6

Difficulty analysis

Questions and published accuracy

Easy	30 Q · 82%
Medium	30 Q · 66%
Hard	15 Q · 41%

Accuracy values are reproduced from the payload.

Mistake distribution

Count by published classification

Concept Error	5
Calculation Error	3
Formula Error	2
NCERT Detail Error	2
Silly Mistake	1
Time Pressure	1
Guess Attempt	0

Source classifications; rows are measured independently.

Subject-wise performance

Subject-wise performance

Accuracy by subject



Physics 88 / 120 Chemistry 80 / 120 Mathematics 76 / 120

Important performance signals

GRADE **A**	RANK **1260**	PERCENTILE **94.2**
BATCH RANK **12**	COHORT SIZE **180**	OVERALL SPEED **4.1**
CONFIDENCE INDEX **87**	READINESS SCORE **78**	HEALTH SCORE **82**
NEGATIVE MARKS **5.25**	GUESS ATTEMPTS **6**	ATTEMPT RATIO **90.7%**
TOTAL QUESTIONS **90**	SUCCESS RATE **70.1%**	BADGE **Top 10% — National**

02 MASTERY MAP

Subject and chapter mastery

Subject marks, accuracy and classification

SUBJECT	MARKS	ACCURACY	RANK	LEVEL	STRONG AREAS	WEAK AREAS
Physics	88 / 120	76%	6	Medium	Mechanics	Thermodynamics and Statistical Mechanics — Advanced Applications in Multi-Stage Reversible Processes with Non-Ideal Corrections and Extended Derivations
Chemistry	80 / 120	71%	9	Easy	Physical Chemistry	Coordination Compounds
Mathematics	76 / 120	63%	14	Hard	Algebra	Definite Integration

Strong, weak and critical chapters by subject

SUBJECT	STRONG CHAPTERS	WEAK CHAPTERS	CRITICAL CHAPTERS
Physics	Thermodynamics and Statistical Mechanics — Advanced Applications in Multi-Stage Reversible Processes with Non-Ideal Corrections and Extended Derivations · Electrostatics · Modern Physics	Ray Optics	—
Chemistry	Physical Chemistry	Coordination Compounds	—
Mathematics	Algebra	Calculus · Probability & Stats	Calculus · Probability & Stats

Chapter mastery · grouped by subject

CHAPTER	ACCURACY	MARKS	TIME	MASTERY
Physics				
Thermodynamics and Statistical Mechanics — Advanced Applications in Multi-Stage Reversible Processes with Non-Ideal Corrections and Extended Derivations	44%	6	4m	Strong
Electrostatics	51%	7	5m	Strong
Modern Physics	58%	8	6m	Strong
Ray Optics	65%	9	7m	Weak

Chapter mastery · grouped by subject · continued

CHAPTER	ACCURACY	MARKS	TIME	MASTERY
Chemistry				
Physical Chemistry	72%	10	8m	Strong
Organic Chemistry	79%	11	9m	Average
Inorganic Chemistry	86%	12	10m	Average
Coordination Compounds	45%	13	4m	Weak
Mathematics				
Algebra	52%	14	5m	Strong
Coordinate Geometry	59%	15	6m	Average
Calculus	66%	16	7m	Critical
Probability & Stats	73%	17	8m	Critical

03 PACE INTELLIGENCE

Time intelligence

Chapter time consumption · measured rows

CHAPTER	SHARE OF TOTAL TIME	TIME CONSUMED	SHARE
Physics			
Thermodynamics and Statistical Mechanics — Advanced Applications in Multi-Stage Reversible Processes with Non-Ideal Corrections and Extended Derivations	5%	4m	5%
Electrostatics	6%	5m	6%
Modern Physics	8%	6m	8%
Ray Optics	9%	7m	9%
Chemistry			
Physical Chemistry	10%	8m	10%
Organic Chemistry	11%	9m	11%
Inorganic Chemistry	13%	10m	13%
Coordination Compounds	5%	4m	5%
Mathematics			
Algebra	6%	5m	6%
Coordinate Geometry	8%	6m	8%

Chapter time consumption · measured rows · continued

CHAPTER	SHARE OF TOTAL TIME	TIME CONSUMED	SHARE
Calculus	9%	7m	9%
Mathematics			
Probability & Stats	10%	8m	10%

Time-management metrics

AVERAGE PER QUESTION
4.1

NAVIGATION COUNT
8

MANAGEMENT SCORE
76

FASTEST QUESTION
Q3 — Mole Concept

FASTEST TIME
0.8

SLOWEST QUESTION
Q58 — Definite Integration

SLOWEST TIME
8.9

Sectional allocation and efficiency

SECTION	ALLOCATED	USED	EFFICIENCY
Physics	60	52	115
Chemistry	55	47	117
Mathematics	65	71	92

04 COMPARISON & TRAJECTORY

Comparison and projection

Comparative performance

Comparison register

BENCHMARK	SCORE	PERCENTILE	GRADE	REFERENCE
Previous test	168	87.2	B	—
Previous month avg	171	88.1	—	—
Batch average	154	82.6	—	—
Institute average	149	80.4	—	—
Top performer	254	99.6	—	Divya Krishnan

VS PREVIOUS TEST	VS PREVIOUS MONTH	VS BATCH AVERAGE	VS INSTITUTE AVERAGE
+14	+11	+28	+33

Projection and readiness

Score trajectory



Expected improvement +4.2% · Risk Low

JEE PERCENTILE	NEET SCORE	EXPECTED AIR
96.2	—	11800

TARGET PROBABILITY	RISK LEVEL	EXPECTED IMPROVEMENT
84	Low	+4.2%

Trajectory values

EXAMINATION / PROJECTION	SCORE
ATS 1	148
ATS 2	158
ATS 3	168
ATS 4 (latest)	182
JEE Main (predicted)	198

READINESS SCORE	CONFIDENCE INDEX	HEALTH SCORE
78	87	82
OVERALL ACCURACY	**SUCCESS RATE**	**BADGE**
73.9%	70.1%	Top 10% — National

05 QUESTION INTELLIGENCE

Question-by-question analysis

14 questions reviewed · published explanations reproduced in full

01 Physics

4 questions

QUESTION 01 PHYSICS

CORRECT

Thermodynamics and Statistical Mechanics — Advanced Applications in Multi-Stage Reversible Processes with Non-Ideal Corrections and Extended Derivations · Mechanics · MCQ · 4 marks · 2m 06s

Derive case 1: $\frac{d}{dx}\left(x^2\sqrt{1-x^2}\right)$ and evaluate $\sum_{k=1}^{3} k^2$.

OPTIONS

A. $$\frac{\pi}{4}$$

B. $\sqrt{3}$ s

C. 4 s

D. A very long alternative answer that must wrap without clipping the question area

STUDENT RESPONSE

A

CORRECT ANSWER

B

PERFORMANCE INSIGHT

Answered correctly — 4 marks · Used u = x^2 instead of u = x^2+1

EXPLANATION

The integral reduces directly:

$$\int_0^1 \frac{x^2}{\sqrt{1-x^2}}\,dx = \frac{\pi}{4}.$$

The substitution $x = \sin\theta$ gives $dx = \cos\theta\,d\theta$ and the limits remain 0 to $\pi/2$. This derivation is intentionally long enough to test multi-line mathematical flow and vector rendering without shortening any source expression.

SOLUTION HINT

Check the sign convention and the limits of integration carefully.

MISTAKE NOTE

Used u = x^2 instead of u = x^2+1

Physics · Q2 · question continued

QUESTION 02 PHYSICS

CORRECT

Non-equilibrium reversible heat transfer with variable coefficients and extended boundary conditions · MCQ · 4 marks · 2m 48s

Derive case 2: $\frac{d}{dx}\left(x^3\sqrt{1-x^2}\right)$ and evaluate $\sum_{k=1}^{4} k^2$.

OPTIONS

A. $$\frac{\pi}{4}$$

B. $\sqrt{3}$ s

C. 4 s

D. A very long alternative answer that must wrap without clipping the question area

STUDENT RESPONSE

B

CORRECT ANSWER

C

PERFORMANCE INSIGHT

Answered correctly — 4 marks · Concept Error

EXPLANATION

Using

$$\sum_{i=1}^{n} i^2 = \frac{n(n+1)(2n+1)}{6},$$

and noting $\alpha^2 + \beta^2 \le \sqrt{\gamma}$, the solution follows. Charge balance for $\mathrm{H_2SO_4}$ and 1.6×10^{-19} C is preserved. Greek symbols, scripts, roots and scientific notation all remain in the source explanation.

02 Chemistry

4 questions

QUESTION 03 CHEMISTRY

CORRECT

Mole Concept · Numerical · 4 marks · 48s

Derive case 3: $\frac{d}{dx}\left(x^4\sqrt{1-x^2}\right)$ and evaluate $\sum_{k=1}^{5} k^2$.

OPTIONS

A. $$\frac{\pi}{4}$$

B. $\sqrt{3}$ s

C. 4 s

D. A very long alternative answer that must wrap without clipping the question area

STUDENT RESPONSE

C

CORRECT ANSWER

D

PERFORMANCE INSIGHT

Answered correctly — 4 marks

EXPLANATION

A multi-line derivation:

EXPRESSION FLAGGED FOR VERIFICATION — SOURCE SHOWN VERBATIM

\begin{aligned} a &= b + c + \frac{1}{2} \\ d &= e + f - \sqrt{\lambda} \\ \therefore\; a+d &= b+c+e+f \\end{aligned}

with $\Delta E = q + w$ and $\nabla \cdot \mathbf{E} = \rho/\varepsilon_0$. The solution continues with $\int_0^\infty e^{-x^2}\,dx = \sqrt{\pi}/2$ and $\lim_{n\to\infty}(1+1/n)^n = e$.

Chemistry · Q5 · question continued

QUESTION 04 CHEMISTRY

INCORRECT

Non-equilibrium reversible heat transfer with variable coefficients and extended boundary conditions · Assertion Reason · 0 marks · 3m 12s

Derive case 4: $\frac{d}{dx}\left(x^5\sqrt{1-x^2}\right)$ and evaluate $\sum_{k=1}^{6} k^2$.

OPTIONS

A. $$\frac{\pi}{4}$$

B. $\sqrt{3}$ s

C. 4 s

D. A very long alternative answer that must wrap without clipping the question area

STUDENT RESPONSE

D

CORRECT ANSWER

A

PERFORMANCE INSIGHT

Answered incorrectly — 0 marks

EXPLANATION

FLAGGED SOURCE — do not correct:

EXPRESSION FLAGGED FOR VERIFICATION — SOURCE SHOWN VERBATIM

```
\frac{1}{2
```

is malformed and must be shown verbatim for verification. The following display expression is also deliberately unbalanced:

EXPRESSION FLAGGED FOR VERIFICATION — SOURCE SHOWN VERBATIM

```
\int_0^1 \frac{x}{\sqrt{1-x^2}
```

. Neither fragment is guessed or silently repaired.

SOLUTION HINT

Check the sign convention and the limits of integration carefully.

03 Mathematics

6 questions

QUESTION 05 MATHEMATICS

CORRECT

Algebra · MCQ · 4 marks · 3m

Derive case 5: $\frac{d}{dx}\left(x^6\sqrt{1-x^2}\right)$ and evaluate $\sum_{k=1}^{7} k^2$.

OPTIONS

A. $$\frac{\pi}{4}$$

B. $\sqrt{3}$ s

C. 4 s

D. A very long alternative answer that must wrap without clipping the question area

STUDENT RESPONSE

A

CORRECT ANSWER

B

PERFORMANCE INSIGHT

Answered correctly — 4 marks · Concept Error

EXPLANATION

The integral reduces directly:

$$\int_0^1 \frac{x^2}{\sqrt{1-x^2}}\,dx = \frac{\pi}{4}.$$

The substitution $x = \sin\theta$ gives $dx = \cos\theta\,d\theta$ and the limits remain 0 to $\pi/2$. This derivation is intentionally long enough to test multi-line mathematical flow and vector rendering without shortening any source expression.

Mathematics · Q14 · question continued

QUESTION 06 MATHEMATICS

INCORRECT

Non-equilibrium reversible heat transfer with variable coefficients and extended boundary conditions · MCQ · 0 marks · 5m 36s

Derive case 6: $\frac{d}{dx}\left(x^7\sqrt{1-x^2}\right)$ and evaluate $\sum_{k=1}^{8} k^2$.

OPTIONS

A. $$\frac{\pi}{4}$$

B. $\sqrt{3}$ s

C. 4 s

D. A very long alternative answer that must wrap without clipping the question area

STUDENT RESPONSE

B

CORRECT ANSWER

C

PERFORMANCE INSIGHT

Answered incorrectly — 0 marks · Used u = x^2 instead of u = x^2+1

EXPLANATION

Using

$$\sum_{i=1}^{n} i^2 = \frac{n(n+1)(2n+1)}{6},$$

and noting $\alpha^2 + \beta^2 \le \sqrt{\gamma}$, the solution follows. Charge balance for $\mathrm{H_2SO_4}$ and 1.6×10^{-19} C is preserved. Greek symbols, scripts, roots and scientific notation all remain in the source explanation.

MISTAKE NOTE

Used u = x^2 instead of u = x^2+1

04 Physics

4 questions

QUESTION 07 PHYSICS

SKIPPED

Ray Optics · MCQ · 0 marks · 4m 24s

Derive case 7: $\frac{d}{dx}\left(x^8\sqrt{1-x^2}\right)$ and evaluate $\sum_{k=1}^{9} k^2$.

OPTIONS

A. $$\frac{\pi}{4}$$

B. $\sqrt{3}$ s

C. 4 s

D. A very long alternative answer that must wrap without clipping the question area

STUDENT RESPONSE

C

CORRECT ANSWER

D

PERFORMANCE INSIGHT

Not attempted — no marks awarded

EXPLANATION

A multi-line derivation:

EXPRESSION FLAGGED FOR VERIFICATION — SOURCE SHOWN VERBATIM

\begin{aligned} a &= b + c + \frac{1}{2} \\ d &= e + f - \sqrt{\lambda} \\ \therefore\; a+d &= b+c+e+f \\end{aligned}

with $\Delta E = q + w$ and $\nabla \cdot \mathbf{E} = \rho/\varepsilon_0$. The solution continues with $\int_0^{\infty} e^{-x^2}\,dx = \sqrt{\pi}/2$ and $\lim_{n\to\infty}(1+1/n)^n = e$.

SOLUTION HINT

Check the sign convention and the limits of integration carefully.

Physics · Q23 · question continued

QUESTION 08 PHYSICS

INCORRECT

Non-equilibrium reversible heat transfer with variable coefficients and extended boundary conditions · Integer · 0 marks · 4m 54s

Derive case 8: $\frac{d}{dx}\left(x^9\sqrt{1-x^2}\right)$ and evaluate $\sum_{k=1}^{10} k^2$.

OPTIONS

A. $$\frac{\pi}{4}$$

B. √3 s

C. 4 s

D. A very long alternative answer that must wrap without clipping the question area

STUDENT RESPONSE

D

CORRECT ANSWER

A

PERFORMANCE INSIGHT

Answered incorrectly — 0 marks · Concept Error

EXPLANATION

FLAGGED SOURCE — do not correct:

EXPRESSION FLAGGED FOR VERIFICATION — SOURCE SHOWN VERBATIM

```
\frac{1}{2
```

is malformed and must be shown verbatim for verification. The following display expression is also deliberately unbalanced:

EXPRESSION FLAGGED FOR VERIFICATION — SOURCE SHOWN VERBATIM

```
\int_0^1 \frac{x}{\sqrt{1-x^2}
```

. Neither fragment is guessed or silently repaired.

05 Chemistry

4 questions

QUESTION 09 CHEMISTRY

INCORRECT

Coordination · MCQ · 0 marks · 3m 48s

Derive case 9: $\frac{d}{dx}\left(x^{10}\sqrt{1-x^2}\right)$ and evaluate $\sum_{k=1}^{11} k^2$.

OPTIONS

A. $$\frac{\pi}{4}$$

B. $\sqrt{3}$ s

C. 4 s

D. A very long alternative answer that must wrap without clipping the question area

STUDENT RESPONSE

A

CORRECT ANSWER

B

PERFORMANCE INSIGHT

Answered incorrectly — 0 marks

EXPLANATION

The integral reduces directly:

$$\int_0^1 \frac{x^2}{\sqrt{1-x^2}}\,dx = \frac{\pi}{4}.$$

The substitution $x = \sin\theta$ gives $dx = \cos\theta\,d\theta$ and the limits remain 0 to $\pi/2$. This derivation is intentionally long enough to test multi-line mathematical flow and vector rendering without shortening any source expression.

06 Mathematics

6 questions

QUESTION 10 MATHEMATICS

CORRECT

Non-equilibrium reversible heat transfer with variable coefficients and extended boundary conditions · MCQ · 4 marks · 4m

Derive case 10: $\frac{d}{dx}\left(x^{11}\sqrt{1-x^2}\right)$ and evaluate $\sum_{k=1}^{12} k^2$.

OPTIONS

A. $$\frac{\pi}{4}$$

B. $\sqrt{3}$ s

C. 4 s

D. A very long alternative answer that must wrap without clipping the question area

STUDENT RESPONSE

B

CORRECT ANSWER

C

PERFORMANCE INSIGHT

Answered correctly — 4 marks

EXPLANATION

Using

$$\sum_{i=1}^{n} i^2 = \frac{n(n+1)(2n+1)}{6},$$

and noting $\alpha^2+\beta^2 \le \sqrt{\gamma}$, the solution follows. Charge balance for $\mathrm{H_2SO_4}$ and 1.6×10^{-19} C is preserved. Greek symbols, scripts, roots and scientific notation all remain in the source explanation.

SOLUTION HINT

Check the sign convention and the limits of integration carefully.

Mathematics · Q47 · question continued

QUESTION 11 MATHEMATICS

INCORRECT

3D Geometry · Numerical · 0 marks · 5m 24s

Derive case 11: $\frac{d}{dx}\left(x^{12}\sqrt{1-x^2}\right)$ and evaluate $\sum_{k=1}^{13} k^2$.

OPTIONS

A. $$\frac{\pi}{4}$$

B. $\sqrt{3}$ s

C. 4 s

D. A very long alternative answer that must wrap without clipping the question area

STUDENT RESPONSE

C

CORRECT ANSWER

D

PERFORMANCE INSIGHT

Answered incorrectly — 0 marks · Concept Error · Used u = x^2 instead of u = x^2+1

EXPLANATION

A multi-line derivation:

EXPRESSION FLAGGED FOR VERIFICATION — SOURCE SHOWN VERBATIM

```
\begin{aligned} a &= b + c + \frac{1}{2} \\ d &= e + f - \sqrt{\lambda} \\ \therefore\; a+d &= b+c+e+f \\end{aligned}
```

with $\Delta E = q + w$ and $\nabla \cdot \mathbf{E} = \rho/\varepsilon_0$. The solution continues with $\int_0^{\infty} e^{-x^2}\,dx = \sqrt{\pi}/2$ and $\lim_{n\to\infty}(1+1/n)^n = e$.

MISTAKE NOTE

Used u = x^2 instead of u = x^2+1

Mathematics · Q58 · question continued

QUESTION 12 MATHEMATICS

SKIPPED

Non-equilibrium reversible heat transfer with variable coefficients and extended boundary conditions · MCQ · 0 marks · 8m 54s

Derive case 12: $\frac{d}{dx}\left(x^{13}\sqrt{1-x^2}\right)$ and evaluate $\sum_{k=1}^{14} k^2$.

OPTIONS

A. $$\frac{\pi}{4}$$

B. √3 s

C. 4 s

D. A very long alternative answer that must wrap without clipping the question area

STUDENT RESPONSE

D

CORRECT ANSWER

A

PERFORMANCE INSIGHT

Not attempted — no marks awarded

EXPLANATION

FLAGGED SOURCE — do not correct:

EXPRESSION FLAGGED FOR VERIFICATION — SOURCE SHOWN VERBATIM

```
\frac{1}{2
```

is malformed and must be shown verbatim for verification. The following display expression is also deliberately unbalanced:

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```
\int_0^1 \frac{x}{\sqrt{1-x^2}
```

. Neither fragment is guessed or silently repaired.

Mathematics · Q63 · question continued

QUESTION 13 MATHEMATICS

INCORRECT

Probability · MCQ · 0 marks · 4m 12s

Derive case 13: $\frac{d}{dx}\left(x^{14}\sqrt{1-x^2}\right)$ and evaluate $\sum_{k=1}^{15} k^2$.

OPTIONS

A. $$\frac{\pi}{4}$$

B. $\sqrt{3}$ s

C. 4 s

D. A very long alternative answer that must wrap without clipping the question area

STUDENT RESPONSE

A

CORRECT ANSWER

B

PERFORMANCE INSIGHT

Answered incorrectly — 0 marks

EXPLANATION

The integral reduces directly:

$$\int_0^1 \frac{x^2}{\sqrt{1-x^2}}\,dx = \frac{\pi}{4}.$$

The substitution $x = \sin\theta$ gives $dx = \cos\theta\,d\theta$ and the limits remain 0 to $\pi/2$. This derivation is intentionally long enough to test multi-line mathematical flow and vector rendering without shortening any source expression.

SOLUTION HINT

Check the sign convention and the limits of integration carefully.

07 Chemistry

4 questions

QUESTION 14 CHEMISTRY

INCORRECT

Non-equilibrium reversible heat transfer with variable coefficients and extended boundary conditions · MCQ · 0 marks · 3m 36s

Derive case 14: $\frac{d}{dx}\left(x^{15}\sqrt{1-x^2}\right)$ and evaluate $\sum_{k=1}^{16} k^2$.

OPTIONS

A. $$\frac{\pi}{4}$$

B. $\sqrt{3}$ s

C. 4 s

D. A very long alternative answer that must wrap without clipping the question area

STUDENT RESPONSE

B

CORRECT ANSWER

C

PERFORMANCE INSIGHT

Answered incorrectly — 0 marks · Concept Error

EXPLANATION

Using

$$\sum_{i=1}^{n} i^2 = \frac{n(n+1)(2n+1)}{6},$$

and noting $\alpha^2+\beta^2 \le \sqrt{\gamma}$, the solution follows. Charge balance for $\mathrm{H_2SO_4}$ and 1.6×10^{-19} C is preserved. Greek symbols, scripts, roots and scientific notation all remain in the source explanation.

06 LEARNING INTELLIGENCE

Learning intelligence

Published revision priorities

Priority revision register

PRIORITY TOPIC	CLASSIFICATION	TIMEFRAME
Critical priority topic 1 — Definite Integration — properties & substitution	Critical	48 hours
High priority topic 2 — Non-equilibrium reversible heat transfer with variable coefficients and extended boundary conditions	High	5 days
Medium priority topic 3 — Definite Integration — properties & substitution	Medium	7 days
Critical priority topic 4 — Non-equilibrium reversible heat transfer with variable coefficients and extended boundary conditions	Critical	10 days
High priority topic 5 — Definite Integration — properties & substitution	High	48 hours
Medium priority topic 6 — Non-equilibrium reversible heat transfer with variable coefficients and extended boundary conditions	Medium	5 days
Critical priority topic 7 — Definite Integration — properties & substitution	Critical	7 days
High priority topic 8 — Non-equilibrium reversible heat transfer with variable coefficients and extended boundary conditions	High	10 days
Medium priority topic 9 — Definite Integration — properties & substitution	Medium	48 hours
Critical priority topic 10 — Non-equilibrium reversible heat transfer with variable coefficients and extended boundary conditions	Critical	5 days
High priority topic 11 — Definite Integration — properties & substitution	High	7 days
Medium priority topic 12 — Non-equilibrium reversible heat transfer with variable coefficients and extended boundary conditions	Medium	10 days

Topic-level accuracy

TOPIC	SUBJECT	ACCURACY	LEVEL
Non-equilibrium reversible heat transfer with variable coefficients and extended boundary conditions	Physics	38%	Critical
Work-Energy-Power	Physics	47%	Weak
Electrostatics — Gauss Law	Physics	56%	Average

Topic-level accuracy · continued

TOPIC	SUBJECT	ACCURACY	LEVEL
Thermodynamics (Physics)	Physics	65%	Strong
Ray Optics — Lenses	Physics	74%	Critical
Semiconductors	Physics	83%	Weak
Mole Concept	Chemistry	92%	Average
Chemical Bonding	Chemistry	43%	Strong
Coordination Compounds	Chemistry	52%	Critical
Definite Integration	Mathematics	61%	Weak
3D Geometry	Mathematics	70%	Average
Probability	Mathematics	79%	Strong
Quadratic Equations	Mathematics	88%	Critical
Matrices & Determinants	Mathematics	39%	Weak

Mistake evidence

QUESTION	SOURCE DETAIL
Q14	Misapplied substitution — used u = x^2 instead of u = x^2+1
Q23	Decimal error in $1/r^2$ computation
Q31	Confused magnetic moment values for d^4 vs d^5
Q47	Wrong distance formula between skew lines
Q52	Sign convention error in lens formula
Q63	Counted 36 outcomes instead of 36 — swapped total
Q71	Rushed mechanism — skipped a step

Weak chapters

NO.	WEAK CHAPTER
1	**Thermodynamics and Statistical Mechanics — Advanced Applications in Multi-Stage Reversible Processes with Non-Ideal Corrections and Extended Derivations**
2	**Calculus (48%)**
3	**Probability & Stats (45%)**
4	**Coordination Compounds (52%)**
5	**Ray Optics (55%)**
6	**Organic Reaction Mechanisms (49%)**

Subject-wise revision sections

Priority topics grouped by subject

PRIORITY TOPIC	CHAPTER	ACCURACY	CLASSIFICATION
Physics			
Non-equilibrium reversible heat transfer with variable coefficients and extended boundary conditions	—	38%	Critical
Work-Energy-Power	—	47%	Weak
Electrostatics — Gauss Law	—	56%	Weak
Mathematics			
Matrices & Determinants	—	39%	Critical
Definite Integration	—	61%	Review
3D Geometry	—	—	Revise
Chemistry			
Chemical Bonding	—	43%	Critical
Coordination Compounds	—	52%	Weak
Also revise			
Ray Optics sign convention	—	—	Revise
Coordination Compounds — NCERT tables	—	—	Revise
Probability — Bayes & distributions	—	—	Revise
Thermodynamic non-ideal corrections	—	—	Revise
Electrostatic boundary conditions	—	—	Revise

Recommended next actions

Published action register

ACTION	RESOURCE	DETAIL
PYQ	**JEE Main 2024 — Definite Integration (5 Qs)**	PYQ Bank
PYQ	**JEE Main 2023 — Ray Optics (4 Qs)**	PYQ Bank
PYQ	**JEE Advanced 2022 — Coordination Compounds**	PYQ Bank
Practice	**20-question drill — Integration by substitution**	20 · Medium
Practice	**10-question drill — Lens formula applications**	10 · Medium
Mock test	**Sectional mock — Mathematics (60 min)**	Sectional
Mock test	**Full syllabus mock — JEE Main pattern**	Full
Lecture	**Integral Calculus — Masterclass (45 min)**	Video
Lecture	**Ray Optics — Sign Convention Simplified**	Video

Weak topics · published list

NO.	WEAK TOPIC
1	Non-equilibrium reversible heat transfer with variable coefficients and extended boundary conditions
2	Definite Integration
3	3D Geometry
4	Ray Optics sign convention
5	Coordination Compounds — NCERT tables
6	Probability — Bayes & distributions
7	Thermodynamic non-ideal corrections
8	Electrostatic boundary conditions

MediXO EduX · Examination Intelligence

End of report · 02 Nov 2026